MIL-C-43062C
24 June 1982
SUPERSEDING
MIL-C-43062B
12 May 1972

MILITARY SPECIFICATION

CLOTH, COATED, NYLON, RESIN MODIFIED

BUTYL COATED, ACID AND FUEL RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document is for one type of cloth, modified butyl rubber coated, in the unvulcanized state. It is to be used for the manufacture of acid and fuel resistant clothing and is vulcanized after assembly of the clothing (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-C-105 - Calcium Chloride, Dihydrate and Calcium Chloride, Anhydrous; Technical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-7254 - Propellants, Nitric Acid.
MIL-C-19377 - Cloth, Twill, Nylon.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-406 - Plastics; Methods of Testing.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

JOINT COMMITTEE ON PRINTING (JCP) PUBLICATIONS

JCP F10 - Manifold Paper, White and Colored

(Copies may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed amoung technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified (see 6.2), at the time of submission of a bid the bidder shall submit to the contracting officer a certified copy of a recent laboratory report covering material which he proposes to deliver. Unless otherwise specified by the contracting officer the bidder shall certify that the material was manufactured in a plant where the coating will be performed if a contract is awarded. This laboratory report shall contain test data which demonstrates that the finished product which the bidder proposes to deliver has been tested in conformance with and found to comply with the requirements of this document. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(a) An independent or commercial laboratory report.
(b) The prospective contractor's own laboratory report.
(c) A governmental laboratory report from a contract within two years of date of submission of bid.

The purpose of the above requirements are to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of the document. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of the document on all deliveries.

3.2 Material.

3.2.1 Base cloth. The base cloth shall be cloth, nylon, twill conforming to MIL-C-19377, except that the cloth shall be undyed.

3.2.2 Coating compound. The compound used for coating the base cloth shall be a composition of resin modified butyl rubber, pigmented to produce a black color and formulated in accordance with the requirements of this document (see 4.3.1.1).

3.2.3 Dusting powder. The dusting powder for application to the coated cloth shall be whiting, talc or other finely divided mineral material which does not support mildew growth (see 4.3.1.1).

3.3 Coated cloth.

3.3.1 Application of coating. The base cloth shall be coated on both sides with the coating compound specified in 3.2.2. Approximately 25 percent of the coating shall be applied to one side of the cloth, and the balance shall be applied to the other side (see 4.3.1.1). After coating, the cloth shall be dusted on both sides with the powder specified in 3.2.3 to produce a uniform dull finish.

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3.3.2 Properties of vulcanized coated cloth. The coated cloth, when vulcanized shall conform to the requirements specified in table I and in 3.3.2.1 through 3.3.2.5 when tested as specified in 4.3.3.

TABLE I. Properties of vulcanized coated cloth

Characteristic	Requirement	
	Minimum	Maximum
Weight, oz/sq. yd.	10.0	11.0
Breaking strength, lbs.		
Warp	80	---
Filling	80	---
Tearing strength, grams		
Warp	960	---
Filling	960	---
Hydrostatic resistance, lbs/sq. in.		
Initial	80	---
After strength of coating	60	---
After cold crack resistance	60	---
Adhesion of coating, lbs/2-inch width	7.0	---
Stiffness, cms, warp only		
At 70°F	---	7.5
At - 20°F	---	8.0
Blocking, scale rating	(1)	
Nitric acid surface tack, rating	II	

3.3.2.1 Abrasion resistance. No visible loose fibers of the base cloth shall be exposed in the center one inch of the abraded portion when tested as specified in 4.3.3. Visible loose fibers shall be regarded as fibers from the nylon yarns used to weave the base cloth and not as entire yarns or yarn plies.

3.3.2.2 Resistance to accelerated weathering. The test specimens shall not become soft and tacky nor stiff and brittle when compared to an unexposed specimen of the same sample unit when tested as specified in 4.3.3.

3.3.2.3 Fuel resistance. There shall be no evidence of cracking, stiffening, flaking or separation of the coating from the base cloth when tested as specified in 4.3.3.

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3.3.2.4 Acid resistance. There shall be no evidence of cracking, stiffening, flaking, separation of the coating from the base cloth or change in color of the indicator paper when tested as specified in 4.3.3.

3.3.2.5 Odor. The vulcanized coated cloth shall be free of objectionable odor when tested as specified in 4.3.3. Odors normally attributable to modified butyl rubber shall not be regarded as objectionable.

3.4 Toxicity clearance. Only those rubber formulations already approved by the appropriate medical service for contact with human skin and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement. Prospective bidders shall send their formulations to the Army Surgeon General for approval (see 6.2).

3.5 Workmanship. The finished cloth shall conform to the quality of product established by this document.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.3.1.1 Certification. Components, materials, and processes listed below shall be accepted on the basis of a contractor's certificate of compliance with the indicated requirements:

a. The coating compound is a composition of resin modified butyl rubber, black in color.
b. The dusting powder is a mineral material that does not support mildew growth.
c. The distribution of coating requirement has been met.

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4.3.2 In-process inspection. Examination shall be made during the coating operation to determine that the coating application is visibly heavier on one side of the coated cloth in accordance with 3.3.1. Whenever nonconformance is noted, correction shall be made to the coated cloth and to the coating operation.

4.3.3 End item testing. The coated cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191, whenever applicable and as listed in table II shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3 continuous yards full width of the cloth vulcanized in the same manner in which the assembled garments are vulcanized, with respect to temperature, curing time and any other controlled processing factor. The lot size shall be expressed in units of 1 linear yard and the sample size (number of sample units) shall be as specified below. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.3.3.1 Standard test conditions. Test results obtained under testing conditions defined in FED-STD-191, FED-STD-406, or FED-STD-601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

TABLE II. End item tests

Characteristic	Requirement reference	Test method
Weight, overall	3.3.2	5041
Breaking strength	3.3.2	5100
Tearing strength	3.3.2	ASTM D 1424
Hydrostatic resistance:		
Initial	3.3.2	5512
After strength of coating	3.3.2	5972 and 5512
After cold crack	3.3.2	5874 1/
Adhesion of coating	3.3.2	5970

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TABLE II. End item tests (cont'd)

Characteristic	Requirement reference	Test method
Stiffness, (warp only):		
At 70°F	3.3.2	5204 2/
At -20°F	3.3.2	5204 2/ 3/
Blocking	3.3.2	5872
Abrasion resistance	3.3.2.1	5304 4/
Resistance to accelerated weathering	3.3.2.2	5804 5/
Chemical resistance:		
Fuel resistance	3.3.2.3	4.4.1.3
Acid resistance	3.3.2.4	4.4.1.4
Nitric acid surface tack	3.3.2	4.4.1.5
Odor	3.3.2.5	4.4.2

1/ The specimens and the instrument shall be conditioned at -40° ± 2°F for a minimum of 4 hours.

2/ The test shall be conducted in still air.

3/ The specimens and the instrument shall be conditioned at -20° ± 2°F for a minimum of 4 hours.

4/ The heavily coated side of the cloth shall be subjected to 40 cycles under a tension of 6 pounds, and under a load of 3 pounds pressure on the test abrader. The abradant shall be "fine" emery cloth. Only three specimens in the warpwise direction shall be tested, and the abrasion resistance shall be determined by visual examination.

5/ The 4 by 6 inch specimen, with the heavily coated side of the cloth facing the light source, shall be exposed in the weatherometer for 100 hours.

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4.4 Methods of inspection.

4.4.1 Chemical resistance.

4.4.1.1 Apparatus and chemicals.

(a) Test jig No. 2 made of aluminum 1100 (see figure 1).
(b) Wax - The wax shall be paraffin or a mixture of paraffin wax and beeswax containing not more than one-third beeswax.
(c) Petroleum jelly.
(d) Aluminum block - Dimensions approximately 4 by 2-7/16 by 1/2 inches.
(e) Liquid vesicant detector paper - M9 chemical agent detector paper (see 6.3).
(f) Propellant nitric acid conforming to type III of MIL-P-7254 (see 6.4).
(g) Glass supporting and cover plates measuring 8 by 8 inches.
(h) Distilled water.
(i) Grade JP-4 fuel conforming to MIL-T-5624.
(j) Onion skin paper conforming to JCP F10 (see 2.1).
(k) Roller - Approximately 5-1/2 inches in diameter, 2 inches wide, and 10 pounds in weight.
(l) Desiccator.
(m) Calcium chloride, anhydrous, conforming to type II, grade A of O-C-105.

4.4.1.2 Preparation of test specimen. The test specimen shall be a section of the coated cloth 6 inches in diameter and shall be prepared as follows:

(a) Place the test specimen between the spacer inserts and seal with melted wax. The wax shall form a continuous bead covering the outer circumferance of the insert assembly.
(b) Place the insert assembly in the test jig sleeve and fill the cavity between the assembly and sleeve wall with melted wax. Mate jig sleeve and jig collar and screw hand tight.
(c) Seal the inside of the sleeve collar with melted wax. The wax seal shall be continuous and shall extend around and out onto the test specimen approximately 1/8 inch above the top of the spacer insert. Turn the jig assembly over and seal the bottom cavity with melted wax in similar fashion.
(d) Place test assembly in desiccator over the calcium chloride and maintain for a minimum of 16 hours.

4.4.1.3 Fuel resistance. The test specimen shall be prepared as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1. Immediately after removal of the assembly from the desiccator, the test specimen shall be covered with 50 mL of JP-4 fuel for 15 minutes at standard conditions. After fuel exposure, the test specimen shall be removed from the test jig, rinsed in distilled water, dried for approximately 4 hours at standard conditions, and examined for requirements in 3.3.2.3.

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4.4.1.4 Acid resistance. The test for acid resistance shall be performed as follows:

(a) Prepare the test specimen as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1.
(b) After conditioning, remove the test assembly from the desiccator and immediately perform the acid exposure test as follows:
  (1) Coat bottom and top edges of the test jig with 1/32 inch of petroleum jelly.
  (2) Place aluminum block in the center of a glass supporting plate with a piece of the vesicant paper, coated side up, on top of the block. Place the test assembly over the block with the heavily coated side of the test specimen up.
  (3) Cover the test specimen with 50 mL of propellant nitric acid and cover the test assembly with a glass cover plate.
(c) After 2 hours acid exposure at standard conditions, remove the test specimen from the test jig, rinse in distilled water, dry at standard conditions for approximately 4 hours, and examine for requirements specified in 3.3.2.4.

4.4.1.5 Nitric acid surface tack. The nitric acid surface tack test shall be performed as follows:

(a) The test specimen shall be prepared as specified in 4.4.1.2 using the apparatus and chemicals specified in 4.4.1.1.
(b) Immediately after removal of the assembly from the desiccator, the test specimen shall be covered with 50 mL of propellant nitric acid for 15 minutes at standard conditions.
(c) After acid exposure, the specimen shall be removed from the test jig, rinsed in distilled water, and dried for approximately 4 hours at standard conditions.
(d) The wax lines shall be removed from the specimen by scraping or trimming. the specimen shall then be placed, exposed side up, on a smooth hard surface and covered with a 2- by 4-inch piece of onion skin paper.
(e) The roller shall be passed over the full length of the paper covered test specimen twice in each direction (four passes) with pressure applied by the weight of the roll only.
(f) After rolling, the onion skin paper shall be peeled from the test specimen and the surface tack determined by comparison with figure 2 and rating standards as specified in 4.4.1.5.1.

4.4.1.5.1 Rating standards for surface tack test.

Rating I - No trace 1/ of coating material on the test paper nor any trace of paper fiber adhering to surface of the test specimen.

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Rating II - A trace 1/ of coating material retained on the test paper or a trace of paper fiber adhering to the test specimen surface.
Rating III - Any degree of coating material or paper fiber transfer in excess of rating II.

NOTE: Figure 2 pictorially represents the Rating Standard.

1/ A trace is defined as coating material or paper fiber transfer covering 25 percent of the test specimen area.

4.4.2 Odor. The examination for objectionable odors shall be made on one specimen, 4 by 6 inches minimum, which has been freely exposed to circulating air at standard conditions for 24 ± 1 hours. The specimen shall fail if it displays odors other than those normally attributable to modified butyl rubber.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended use. The coated cloth is intended for the manufacture of protective clothing which is resistant to rocket fuels and oxidizers and suitable for use at low temperatures. the protective clothing shall be assembled using unvulcanized coated cloth and vulcanized after assembly of the clothing.

6.2 Ordering data. Acquisition documents for the protective clothing should specify the following:

(a) Title, number and date of this document.
(b) When laboratory report approval is required (see 3.1).
(c) Whether formula approval has been obtained (see 3.4).

6.3 Liquid vesicant detector paper. Liquid vesicant detector paper "M9" can be procured under Chemical Systems Laboratory Purchase Description "EA-P-1162 - Paper, Chemical Agent Detector: M9" dated 13 June 1980, prepared by the US Army Armament Research and Development Command, Aberdeen Proving Ground, MD 21010, and is available through the contracting activity (see 4.4.4.1).

6.4 Nitric acid. Propellant nitric acid conforming to type III of MIL-P-7254 is available from General Chemical Division, Allied Chemical and Dye Corp., 12 South 12th Street, Philadelphia, PA (see 4.4.4.1).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - SH
DLA - CT

User activities:

Army - AR
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Preparing activity:

Army - GL

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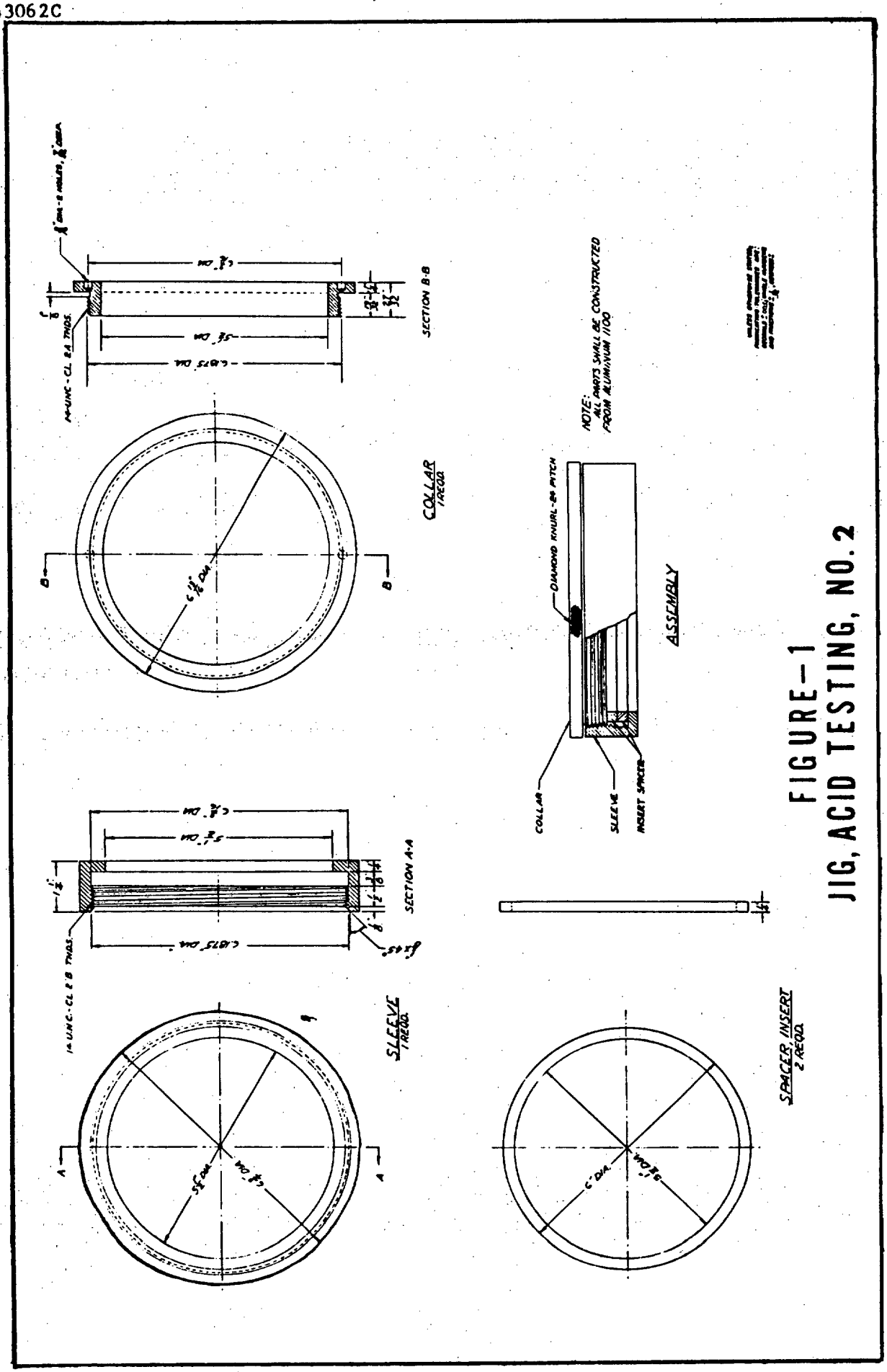


FIGURE-1
JIG, ACID TESTING, NO. 2

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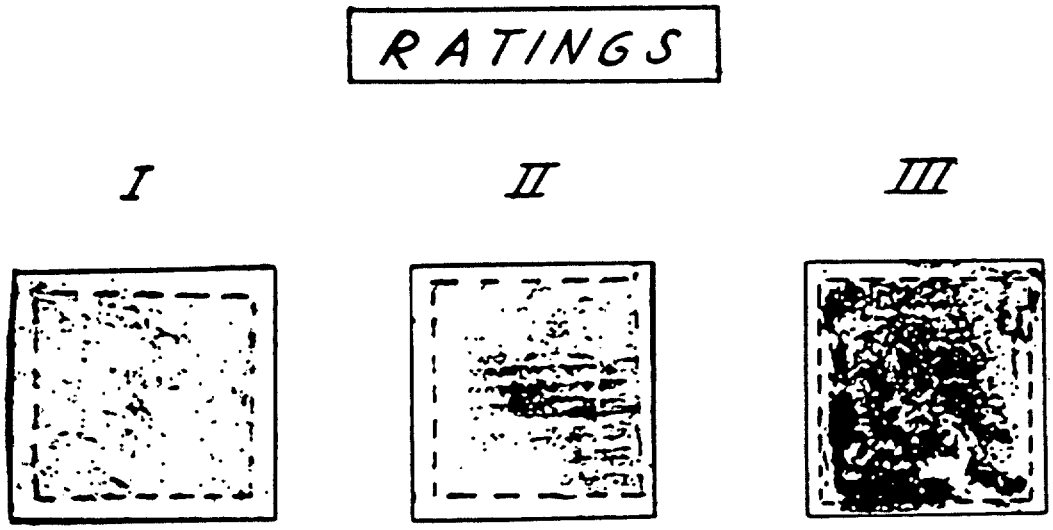


FIGURE-2. SURFACE TACK STANDARDS
NITRIC ACID TEST

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